Unit 01: Getting Started with Linux

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Objectives

After studying this unit, you will be able to

- understand the operating system
- know the history of Linux
- understand the features of Linux
- understand the basic commands of Linux
- understand the shell of Linux

Introduction

An operating system is the low-level software that schedules tasks, allocates storage, and handles the interfaces to peripheral hardware, such as printers, disk drives, the screen, keyboard, and mouse. An operating system has two main parts: the kernel and the system programs.

- The kernel allocates machine resources—including memory, disk space, and CPU cycles—to all other programs that run on the computer.
- The system programs include device drivers, libraries, utility programs, shells (command interpreters), configuration scripts and files, application programs, servers, and documentation.

The Linux kernel was developed by Linus Torvalds. He released Linux version 0.01 in September 1991. The name 'Linux' is a combination of Linus and UNIX. The Linux OS, is a product of the Internet and is a free OS. Linux is free software. "Free software" is a matter of liberty, not price. To understand the concept, you should think of "free" as in "free speech," not as in "free beer." The UNIX OS is the common ancestor of Linux. A Linux distribution comprises the Linux kernel, utilities, and application programs. Many distributions are available, including Ubuntu, Fedora, Red Hat, Mint, OpenSUSE, Mandriva, CentOS, and Debian.

1.1 The History of UNIX and GNU–Linux

The Heritage of Linux: UNIX

The UNIX system was developed by researchers who needed a set of modern computing tools to help them with their projects. When the UNIX OS became widely available in 1975, Bell Labs offered it to educational institutions at nominal cost. The schools, colleges, universities and industries accepted the OS and worked on it.

Linux-BSD

One version is called the Berkeley Software Distribution (BSD) of the UNIX system (or just Berkeley UNIX).

Fade to 1983

Richard Stallman announced the GNU Project for creating an operating system, both kernel and system programs. GNU, which stands for Gnu's Not UNIX, is the name for the complete UNIX-compatible software system.

Next Scene, 1991

The GNU Project has moved well along toward its goal. Much of the GNU operating system, except for the kernel, is complete.

The Code is Free

The tradition of free software dates back to the days when UNIX was released to universities at nominal cost, which contributed to its portability and success. This tradition eventually died as UNIX was commercialized. Another problem with the commercial versions of UNIX related to their complexity.

1.2 What is so good about Linux

- Standards
- Applications
- Peripherals
- Software
- Platforms
- Emulators
- Virtual Machines
- Xen
- VMWare
- KVM
- Qemu
- Virtual Box

Standards

In 1985, the POSIX standard was developed, which is based largely on the SVID and other earlier standardization efforts. These efforts were spurred by the U.S. government, which needed a standard computing environment to minimize its training and procurement costs.

Applications

A rich selection of applications is available for Linux—both free and commercial—as well as a wide variety of tools: graphical, word processing, networking, security, administration, Web server, and many others.

Peripherals

Linux often supports a peripheral or interface card before any company does. Unfortunately some types of peripherals—particularly proprietary graphics cards—lag in their support.

Software's

Also important to users is the amount of software that is available—not just source code but also prebuilt binaries that are easy to install and ready to run. These programs include more than free software.

Platforms

Linux is not just for Intel-based platforms: It has been ported to and runs on the PowerPC. Nor is Linux just for single-processor machines.

Emulators

Linux supports programs, called emulators, that run code intended for other operating systems.

Virtual Machines

A virtual machine appears to the user and to the software running on it as a complete physical machine. The software that provides the virtualization is called a virtual machine monitors (VMM) or hypervisor. Each VM can run a different OS from the other VMs.

Xen

Xen, which was created at the University of Cambridge and is now being developed in the open-source community, is an open-source VMM. Xen introduces minimal performance overhead when compared with running each of the operating systems natively.

VMWare

VMware, Inc. offers VMware Server, a free, downloadable, proprietary product you can install and run as an application under Linux. VMware Server enables you to install several VMs, each running a different OS, including Windows and Linux.

KVM

The Kernel-based Virtual Machine (KVM) is an open-source VM and runs as part of the Linux kernel.

Qemu

Qemu, is an open-source VMM that runs as a user application with no CPU requirements. It can run code written for a different CPU than that of the host machine.

Virtual Box

Virtual Box is an open-source VM developed by Sun Microsystems.

1.3 Why Linux Is Popular with Hardware Companies and Developers

Two trends in the computer industry set the stage for the growing popularity of UNIX and Linux. These are proprietary and generic operating systems.

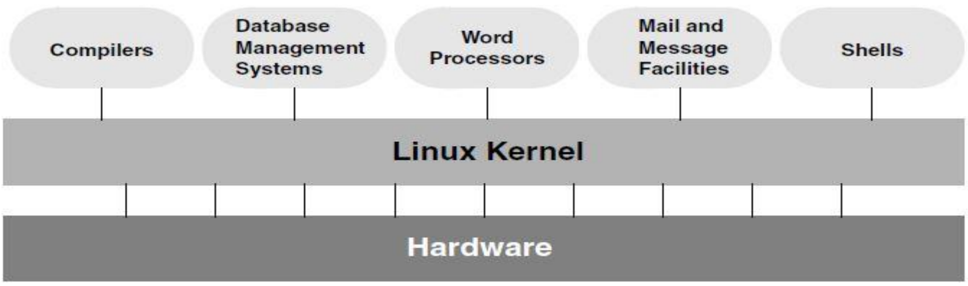
Proprietary operating systems: A proprietary OS is one that is written and owned by the manufacturer of the hardware (for example, DEC/Compaq owns VMS). Today's manufacturers need a generic OS that they can easily adapt to their machines.

Generic operating systems: Generic OS is written outside of the company manufacturing the hardware and is sold (UNIX, OS X, Windows) or given (Linux) to the manufacturer. Linux is a generic OS.

Linux emerged to serve both needs: It is a generic OS that takes advantage of available hardware. A portable OS is one that can run on many different machines. More than 95% of the Linux OS is written in the C programming language. Because Linux is portable, it can be adapted (ported) to different machines and can meet special requirements. The ancestor of Linux is UNIX. The UNIX OS was written in assembly language. For this reason, the original UNIX OS was not portable. To make UNIX portable, Thompson developed the B programming language, a machine-independent language. Dennis Ritchie developed the C programming language by modifying B and, with Thompson, rewrote UNIX in C in 1973.

1.4 Overview of Linux

Like UNIX, it is also a well-thought-out family of utility programs and a set of tools that allow users to connect and use these utilities to build systems and applications.



Kernel Programming Interface

The Linux kernel—the heart of the Linux OS—is responsible for allocating the computer's resources and scheduling user jobs so each one gets its fair share of system resources. Programs interact with the kernel through system calls. A programmer can use a single system call to interact with many kinds of devices. For example, there is one write() system call, rather than many device-specific ones. It also makes it possible to move programs to new versions of the OS without rewriting them (provided the new version recognizes the same system calls).

Supports Many Users

Depending on the hardware and the types of tasks the computer performs, a Linux system can support from 1 to more than 1,000 users, each concurrently running a different set of programs.

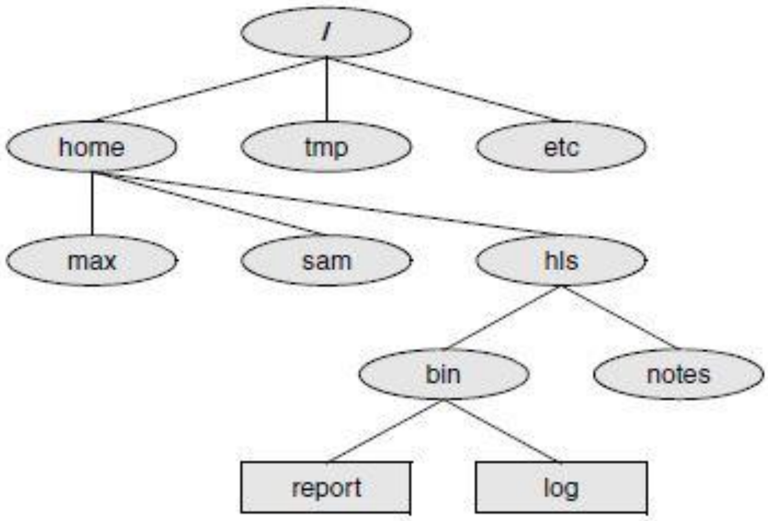
Runs Many Tasks

Linux is a fully protected multitasking OS, allowing each user to run more than one job at a time. Processes can communicate with one another but remain fully protected from one another, just as the kernel remains protected from all processes.

Secure Hierarchical File system

The Linux file system provides a structure whereby files are arranged under directories, which are like folders or boxes. Each directory has a name and can hold other files and directories. Directories, in turn, are arranged under other directories, and so forth, in a treelike organization.

Linux File System Structure



Standards

The Linux File system Standard (FSSTND) was developed, which has since evolved into the Linux File system Hierarchy Standard (FHS).

Links

A link allows a given file to be accessed by means of two or more names. The alternative names can be located in the same directory as the original file or in another directory.

Security

Like most multiuser OSs, Linux allows users to protect their data from access by other users. It also allows users to share selected data and programs with certain other users by means of a simple but effective protection scheme.

The Shell

The shell can work as a command interpreter as well as a programming language.

The shell as a command interpreter: In a textual environment, the shell—the command interpreter—acts as an interface between you and the OS. A number of shells are available for Linux. The four most popular shells are:

- Bourne Again Shell
- Debian Almquist Shell
- TC Shell
- Z Shell

The shell as a Programming Language: The shell is a high-level programming language. Shell commands can be arranged in a file for later execution which is known as shell scripts. This flexibility allows users to perform complex operations with relative ease.

Filename Generation

When you type commands to be processed by the shell, you can construct patterns using characters that have special meanings to the shell. These characters are called wildcard characters. The patterns, which are called ambiguous file references, are a kind of shorthand.

Completion

In conjunction with the Readline library, the shell performs command, filename, pathname, and variable completion.

Device-Independent Input and Output

- Redirection: When you give a command to the Linux OS, you can instruct it to send the output to any one of several devices or files. This diversion is called output redirection.
- Device independence: In a similar manner, a program's input, which normally comes from a keyboard, can be redirected so that it comes from a disk file instead.

Shell Functions

Many shells, including the BASH, support shell functions that the shell holds in memory so it does not have to read them from the disk each time you execute them.

Job Control

Job control is a shell feature that allows users to work on several jobs at once, switching back and forth between them as desired.

A Large Collection of Useful Utilities

Linux includes a family of several hundred utility programs, often referred to as commands. These utilities perform functions that are universally required by users. For example: Sort utility.

Inter process Communication

Linux enables users to establish both pipes and filters on the command line. A pipe sends the output of one program to another program as input. A filter is a special kind of pipe that processes a stream of input data to yield a stream of output data.

System Administration

On a Linux system the system administrator is frequently the owner and only user of the system. This person has many responsibilities. The first responsibility may be to set up the system, install the software, and possibly edit configuration files.

Additional Features of Linux

The developers of Linux included features from BSD, System V, and Sun Microsystems' Solaris, as well as new features, in their OS.

GUIs: Graphical User Interfaces

X11:

Given a terminal or workstation screen that supports X, a user can interact with the computer through multiple windows on the screen, display graphical information, or use special-purpose applications to draw pictures, monitor processes, or preview formatted output. Usually two layers run on top of X: a desktop manager and a window manager.

Desktop manager: A desktop manager is a picture-oriented user interface that enables you to interact with system programs by manipulating icons instead of typing the corresponding commands to a shell.

Window manager: A window manager is a program that runs under the desktop manager and allows you to open and close windows, run programs, and set up a mouse so it has different effects depending on how and where you click.

(Inter)Networking Utilities

Linux network support includes many utilities that enable you to access remote systems over a variety of networks. In addition to sending email to users on other systems, you can access files on disks mounted on other computers.

Software Development

One of Linux's most impressive strengths is its rich software development environment. Linux supports compilers and interpreters for many computer languages.

Utilities

These utilities facilitate you for a large task. There are various utilities like ls, cd, cat, mv, cp, echo and date.

Summary

- An operating system has two main parts: the kernel and the system programs.
- A Linux distribution comprises the Linux kernel, utilities, and application programs.
- Linux supports programs, called emulators, that run code intended for other operating systems.
- The software that provides the virtualization is called a virtual machine monitor.
- The Kernel-based Virtual Machine (KVM) is an open-source VM and runs as part of the Linux kernel.
- A portable OS is one that can run on many different machines.
- The four most popular shells are: Bourne Again Shell, Debian Almquist Shell, TC Shell and Z Shell.
- Shell commands can be arranged in a file for later execution which are known as shell scripts.

Keywords

- **ls utility**: A utility used for listing the contents of a directory.
- **mv utility:** A utility used for moving the files from one directory to another.
- **cd utility:** A utility used for changing the directory.
- **Operating System:** An operating system is the low-level software that schedules tasks, allocates storage, and handles the interfaces to peripheral hardware, such as printers, disk drives, the screen, keyboard, and mouse.
- **Emulators**: Linux supports programs, called emulators, that run code intended for other operating systems.
- **Proprietary OS:** It is the one that is written and owned by the manufacturer of the hardware.
- **Generic OS:** It is written outside of the company manufacturing the hardware and is sold or given to the manufacturer.

Self Assessment

1. An operating system is responsible for
   A. Scheduling the tasks.
   B. Allocation the storage
   C. Handling the interfaces for peripherals
   D. All of the above
2. An operating system has
   A. Kernel
   B. System programs
   C. Both of the above
   D. None of the above
3. Which of these are distributions of Linux?
   A. Fedora
   B. RedHat
   C. CentOS
   D. All of the above
4. A Linux distribution comprises of
   A. Application programs
   B. Utilities
   C. Kernel
   D. All of the above
5. Which of these shells are available in Linux?
   A. TC Shell
   B. Z Shell
   C. Debian Almquist Shell
   D. All of the above
6. Which is the core of operating system?
   A. Kernel
   B. Commands
   C. Shell
   D. Script

7. Which of these utilities is used to show which files and folders are available in the system?
    A. ls
    B. cat
    C. rm
    D. All of the above
8. Linux OS supports
    A. Multi User
    B. Multi Process
    C. Multi-tasking
    D. All of the above
9. Which of these utilities is used for concatenation of files?
    A. ls
    B. cat
    C. rm
    D. echo
10. Which of these utilities is used for displaying the contents of a file?
    A. ls
    B. cat
    C. rm
    D. echo
11. Linux is an example of
    A. Web browser
    B. Word processing software
    C. Operating system
    D. Photo editor
12. Who founded the Linux Kernel?
    A. Richard Stallman
    B. Ben Thomas
    C. Linus Torvalds
    D. None of the above
13. Which of the following OS is not based upon Linux?
    A. BSD
    B. Redhat
    C. Ubuntu
    D. CentOS
14. What is available to users in Linux OS?
    A. Source code
    B. Prebuilt binaries
    C. Both source code and prebuilt binaries
    D. None of the above

15. Which of these represents system programs?

    A. Libraries
    B. Device drivers
    C. Servers
    D. All of the above

Answers for Self Assessment

1.	D	2.	C	3.	D	4.	D	5.	D
6.	A	7.	A	8.	D	9.	B	10.	B
11.	C	12.	C	13.	A	14.	C	15.	D

Review Questions:

1. What is an operating system? Explain its main parts.
2. What are the features of Linux? Explain.
3. What are proprietary and generic operating systems? Explain why Linux is popular with hardware companies and developers?
4. What is so good about Linux? Explain about its applications, peripherals, platforms and standards.
5. What is kernel programming interface? Explain.
6. What are the basic utilities in Linux? Explain.

Further Readings

Mark G Sobell, A Practical Guide to Linux Commands, Editors, and Shell Programming, Second Edition.

Web Links

https://www.redhat.com/en/topics/linux/what-is-linux